

KUO, CHEN CHIH 郭晨志

M.S. in Chemical Engineering • Licensed environmental specialist

Seeking EHS Engineer

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Xinzhuang, New Taipei • Open to relocation / shifts

Profile

M.S. candidate in Chemical Engineering at NTUST, holding three environmental specialist licences — **wastewater, waste disposal and indoor air quality** — with **occupational safety & health** exams scheduled; hands-on in lab equipment & safety management and **thermal hazard analysis (TGA / DSC)**.

3	NT\$ 2M+	6	3 +2
Battery-analysis techniques run in the industry project: EIS, OCV drip, X-ray CT	Precision equipment procured & accepted	Labs across 3 departments served with TGA testing	3 environmental licences held, 2 OSH exams scheduled

Experience

2025 – Present	Sustainable Electrochemical Energy Development (SEED) Center, NTUST Lab Equipment Procurement & Asset Maintenance Manager Procurement (>NTD 2M): Led procurement of glovebox O₂ sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes and low-temperature baths. TGA testing service: Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST. Instrument maintenance: Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500. - Maintained close, responsive contact with vendor engineers (Waters, TA Instruments, Uni-Onward), frequently securing on-site repair within 24 hours of a fault report — significantly reducing instrument downtime.
Summer 2022	New Taipei City Dept. of Urban Development Summer Intern, Housing Development Section Reviewed rental subsidy applications and indexed confidential records in government databases; handled frontline public inquiries, building strong familiarity with administrative workflows and SOPs.
2021 – 2024	Tunghai University Student Housing / GEPT Resident Assistant (2 yrs) / Exam Staff (part-time) Oversaw safety and daily order in a 100-resident dormitory, mediating resident disputes and liaising with university administration; as part-time exam staff, handled test-paper check-in with zero errors.

Research & Industry

Long-Life Li Battery Evaluation for Smart Water Meters

Partner: a smart-water-meter battery maker **Project lead**

Screening methodology Designed full-stage test protocols, experimental workflows and statistical validation models.	Advanced analysis EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis.
Database & SOP Established the test database and standardized high-throughput screening procedures (SOP).	Training & tech transfer Trained QA and R&D teams to independently run large-batch screening and data interpretation.

Nature Energy Journal submission — battery thermal stability study Status Under review
Co-author Responsible for TGA thermal measurement
Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions.

Conferences & Training

Presenter 2026 TwICHE Annual Meeting Title: <i>Weakly Solvating Allyl Ether Electrolytes for Subzero Li SPAN</i> Batteries: Solvation Structure, Transport Properties, and Separator Wettability
Vendor seminar TA Instruments ARES-G3 launch & applications forum 2026 / 07 / 17
Vendor lecture TA Instruments TGA lecture, Taiwan 2026 / 06 / 03

EHS Licenses

✓	Dedicated Wastewater and Sewage Treatment Specialist (Class B) Certified 2026
✓	Indoor Air Quality Maintenance Management Specialist Certified 2026
✓	Waste Disposal Technician (Class A) Certified 2026
○	Occupational Safety and Health Management Personnel (Level B) 2026 / 11 • mid-Nov exam scheduled
○	Occupational Safety Management Specialist (Level A) 2027 / 03 • exam scheduled

Education

2025 – 2027 (expected)	National Taiwan University of Science and Technology M.S. in Chemical Engineering Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection.
2020 – 2024	Tunghai University B.S. in Chemical & Materials Engineering Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).

Skills & Instruments

Electrochemistry & batteries - Impedance spectroscopy (EIS) - Battery aging analysis - Reliability analysis - Low-temp electrolytes
Characterization - Thermal analysis (TGA / DSC) - Synchrotron (TLS20A / TPS27A / 45A) - Rheology (ARES-G3) - Viscometer & densimeter (DMA5002)
Process & operations - Lithography (nanoimprint / self-assembly) - Equipment procurement & maintenance

Software

Aspen Plus V14	Aspen Dynamics
Microsoft Office	OriginPro
TA Universal Analysis	TA TRIOS

Instruments

TGA Q500 Maintenance
DSC 2500 Maintenance
ARES-G3 Operation
Glovebox O ₂ sensor Procurement
DSC sample press Procurement + service
Hydraulic electrode punch Procurement
Vacuum pump Procurement
Eppendorf Pipette Procurement
Low-temperature bath Procurement